

Airframe Technology Master

Lectures 06.

General Structures Design Principles: Mechanical Joints Part I

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Introduction

Parameters to be evaluated in order to define a joint:

- Structural Integrity: Safe Life, Fail Safe and Damage Tolerant structure
- Maintainability: Permanent, removable and usually removable joints
- Loads transferred
- Tolerances and manufacturing process
- Fasteners installation
- Corrosion prevention and electrical bonding

Continuous joint design criteria

- Typical distances: edge distance and pitch
- Assembly criteria
- General criteria
- Selection Criteria summary

This lesson is intended to provide knowledge about the main types of joining methods used at aerospace sector, the parameters that should be taken into account before defining the configuration of a joint and the main criteria to take into account in the detail design of the joint.

The joints are usually classified in permanent and removable joints but there are much more parameters to be taken into account in a joint, we analyze these parameters in the next paragraph.

Structure Integrity

According to CS/FAR 25.571, in order to guarantee the structural integrity of the A/C structure, there are several certification approaches:

- Safe Life structure
- Fail Safe structure
- Damage tolerant structure

Safe Life Structure: It is calculated the fatigue life of an structure, therefore it is supposed that the structure withstands the loads during its service life without detectable cracks. Once the structure reaches its service life it must be replaced by a new one. The Landing Gear usually requires this kind of certification approach.

Fail Safe Structure: Structure where the load paths are duplicated therefore a single failure never leads to a Catastrophic failure

Damage Tolerant Structure: Is that structure designed with the assumption that it will suffer fatigue, corrosion or accidental damage so a crack will appear, the critical elements are identified and analyzed, residual strength must be calculated considering the cracks.

Maintainability

The maintenance task and the time required to perform them is quite important for the customers, because they want the aircraft to be permanently available to fly. In order to allow the maintenance task it is necessary to get access to many aircraft areas. Depending of the frequency of the inspections the structures which have to be removed can be attached as:

- Permanent Joint
- Removable Joint
- Usually removable Joint

Maintainability

Permanent Joint:

Joints which are not expected to be removed during the life of the aircraft.

Bonding or fasteners like Rivets or Hi-lok are used, this name does not mean that this joint can not be removed, the rivets/hi-lok can be removed and so do most of adhesives, but the removal usually affects the structure requiring a rework of the structure and the fasteners can not be used again.

Maintainability

Permanent Joint:

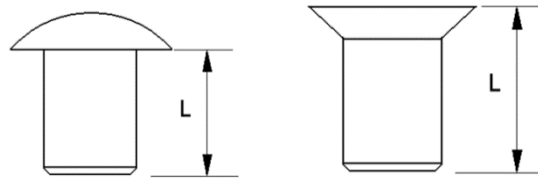
Solid Rivets:

One head is already formed from the standard manufacturer and the other end is cylindrical and is conformed at assembly with and specific tooling.

The rivets head can be protruding or countersunk, at one or at both ends.

Typical materials are Aluminium, Titanium or CRES but there are also Monel rivets.

These joints are able to transfer mainly shear loads.

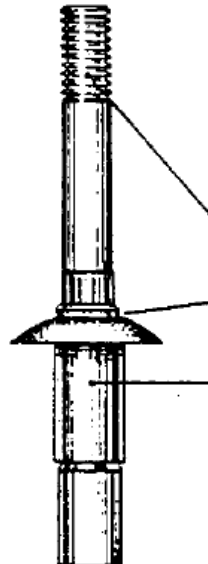


Maintainability

Permanent Joint:

Blind rivets:

Rivets used when only there is access to one side of the structure so solid rivets can not be used.



Maintainability

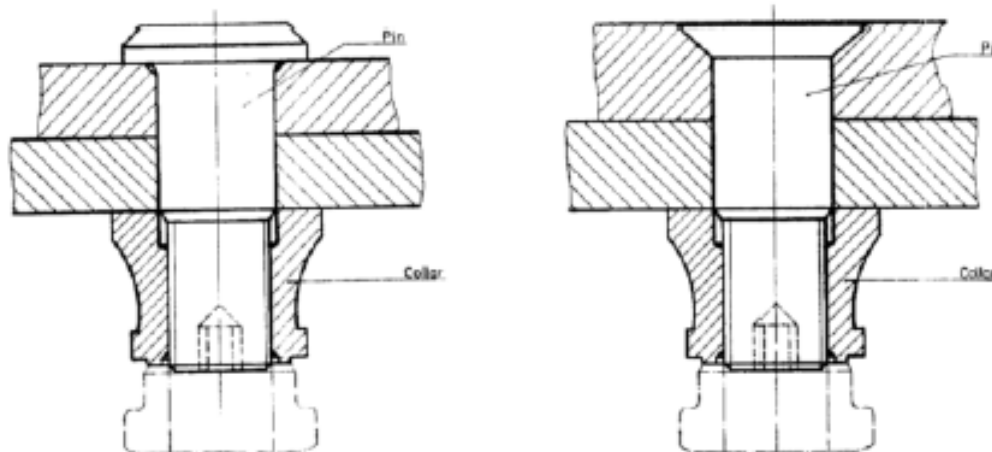
Permanent Joint:

Hi-loks / Hi-lites:

They are similar to a bolt/nut joint but their installation process is different.

The nuts are special standard parts called collar with two different zones, the threaded nut which engages the hi-lok and an hexagonal head where the torque tighten is applied, these two parts junction breaks when the torque applied reaches a defined value .

They are able to transfer traction but also shear because they are installed with an interference fit.



Maintainability

Permanent Joint:

Hi-loks / Hi-lites:

The pin can be made of Titanium or CRESS, we use one or the other depending on the weight and load requirements, CRESS hi-loks have higher shear and tension allowable loads than Titanium.

The collars are made of CRES or Aluminium, the CRES ones are used when higher tension load allowables are required.

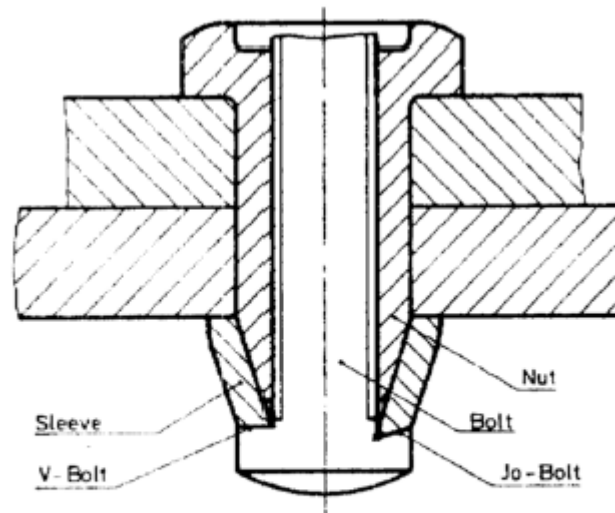
The CRES and titanium are galvanically not compatible with aluminium parts so they should be used with a surface protection which avoids the aluminum part corrosion, for example IVD (aluminium Ion Vapour Deposition).

Maintainability

Permanent Joint:

Jo-bolts or blind bolts:

Are fasteners which can be installed at low accessibility areas just like the blind rivets, but the closing head geometry is more accurate so they are able to transfer not only shear loads but also tension loads.



Remache ciego Jo-Bolt

Maintainability

Permanent Joint:

Stud pin:

Special kind of fastener with a threaded shank at both ends and a flange in the middle. At one end two parts can be permanently joined closing by a collar and at the other end a part can be joined with an standard nut.

Maintainability

Removable Joint:

Joints which are expected to be removed during the life of the aircraft.

This type of joints commonly uses bolts which can be removed and reinstalled several times without affecting the structure or the fasteners itself .

These joints are used when there are maintainability or inspection requirements that require access to an specific area through an access panel or for installing equipment or for parts which can require to be substituted due to wear, erosion, etc.

This kind of joint is considered to transfer only tension loads, no shear, the hole diameter is several millimeters larger than the bolt diameter, so there is no contact between the bolt shank and the hole.

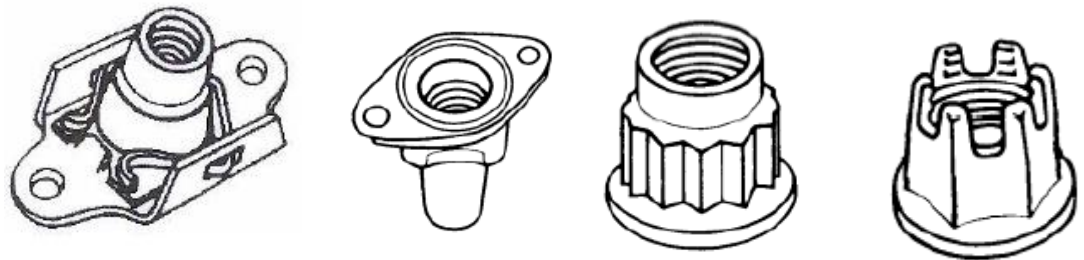
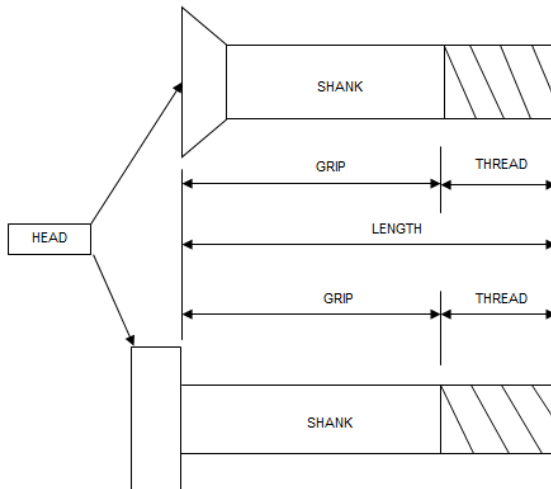
The only shear loads are transferred through the friction due to the torque tighten. High loads can overcome the friction and generate a displacement until the bolts get in contact with the holes, but not all the bolts will get in contact with the holes therefore only a percentage of the bolts is considered to transfer load.

Maintainability

Removable Joint:

Bolts are usually installed with anchor-nuts in order to avoid the missing of the nut during maintenance task or during the A/C assembly.

Bolt typically used is NAS6703-19, material CRES A286.



Maintainability

Usually removable joint:

This is a joint which has to be uninstalled quite often so special quick release fasteners are commonly used, and special sealing process is defined in order to keep the sealing at install/uninstall operations without the need of perform a new sealing.

Access types in function of the frequency of access:

TASK	EQUIPEMENT	ACCESS REQUIRED
Servicing and Daily Activities		Access possible without tools e.g. Quick-release latches. Hinged Doors/panels.
Line maintenance		Access possible through doors/panels fitted with captive type quick release fasteners. Doors/panels remain attached to aircraft.
Heavy maintenance		Access possible through doors/panels fitted with captive type quick release fasteners, or if necessary, screws. All screws in the same panel to be identical.
Unscheduled maintenance	MMEL ⁸ Components LRUs/Components with low MTBUR	Access possible through doors/panels fitted with captive type quick release fasteners, or if necessary, screws. Doors/panels remain attached to aircraft.
	Other Components or Equipment	Access possible through doors/panels fitted with captive type quick release fasteners.

Maintainability

Recommended dimensions for access holes:

<i>ACCESS TYPE</i>	<i>SIZE</i>
Whole body external top bottom access	(310 x 460) mm
Whole body internal top bottom access	(350 x 500) mm
Whole body external side access	(400 x 460) mm
Whole body internal side access	(400 x 500) mm
Head and arms external top bottom access	(310 x 460) mm
Head and arms internal top bottom access	(310 x 460) mm
Head and arms external side access	(310 x 460) mm
Head and arms internal side access	(310 x 460) mm
Head and arm	(200 x 300) mm
2 Arms (Up to 300 mm)	(125 x 250) mm
Arm + Sight	(125 x 200) mm
Arm	(125 x 125) mm
Endoscope (Above 200 mm)	(15 x 15) mm
Endoscope (Up to 200 mm)	(10 x 10) mm

Reparability

Permanent fasteners joints , in Primary Structure, should allow to be replaced at least two times in case of any repair is needed.

Solid and blind rivets can be replaced by a rivet with the same diameter, as the shank expands at installation and gets contact with the hole.

Solid rivets can be replaced by hi-loks, but then an oversize fastener shall be used.

Hi-loks and hi-lites have to be replaced by a higher diameter fastener, there are oversize fasteners, whose diameter is 0,4 mm larger (1st oversize) for first replacement and 0,8mm larger (2nd oversize) for second replacement of the fasteners.

Loads transferred

The joint has to withstand and transfer the static, dynamic and thermal loads from static strength point of view and from Fatigue & Damage Tolerance point of view.

The loads at the fasteners can be tension loads or shear loads, bending moments and torsion moments introduce additional tension or shear loads at the fasteners respectively.

Each fastener type and its installation conditions is more adequate to carry a type of load.

Loads transferred

Tension loads: Bolts should be used mainly for tension loads transfer, they are not a good option for shear loads because the bolt are usually installed with high clearance into the holes, so there is no direct contact between the bolt pin and the hole, for low shear loads the tightening torque creates a compression between fasteners heads and the joined parts so friction is able to transfer the load.

Shear loads: Rivets, due to shank expansion at installation, have interference with the structure hole, so the are able to transfer the shear load. The rivets closing head geometry is not controlled with high precision since it is conformed during installation so the tension load which they can carry is not guaranteed, therefore rivets should not be used for transfer tension loads.

There are special fasteners which can withstand both shear and tension loads, p.e. hi-lok, hi-lite, jo-bolts, lock-bolt, etc

Tolerances and manufacturing process

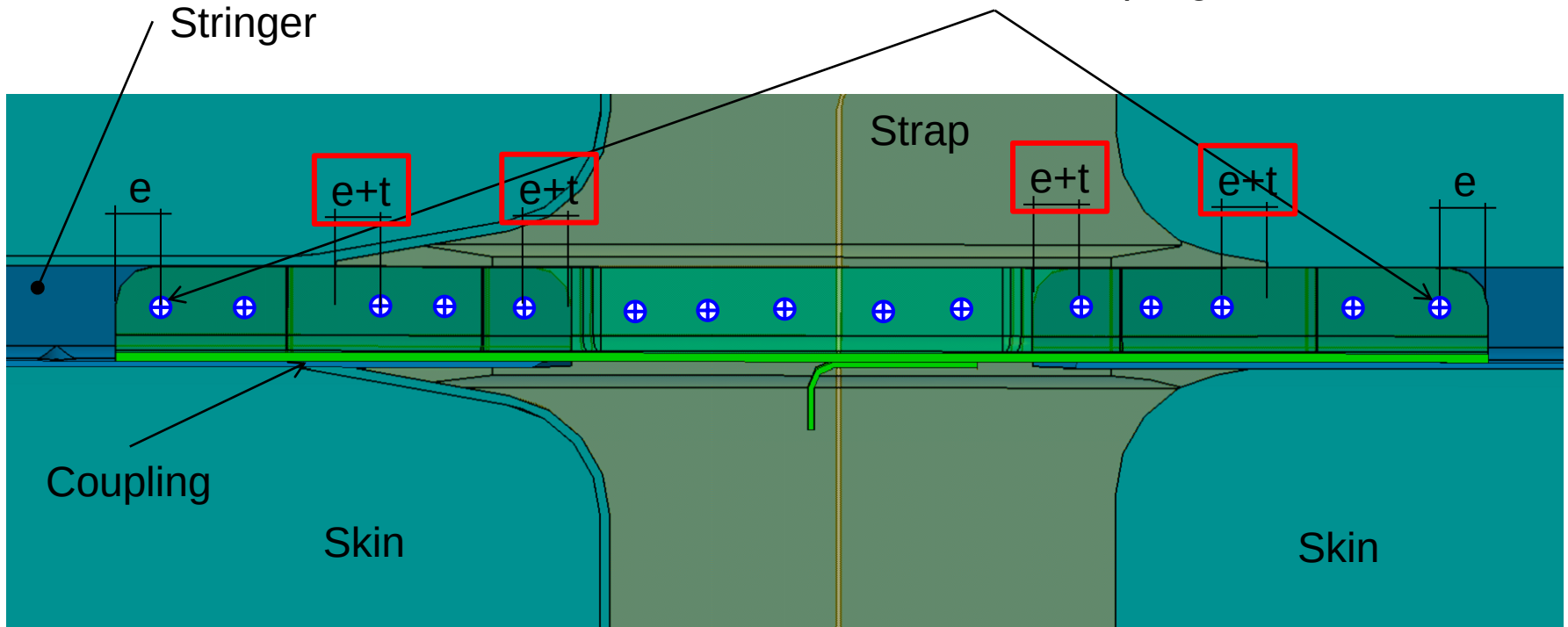
Due to the parts manufacturing tolerances and the assembly tolerances, each joint will have to cope with the tolerances stack up of the parts or sub-assemblies to be joined.

The joint design is completely linked to the assembly process, so it will have to be agreed with production according to the manufacturing process and the manufacturing tooling.

A joint can be defined in such a way that the tolerances stack up is completely absorbed by the joint in order to minimize the tolerances transferred to the next structure integration step:

- Permanent joints: This can be achieved providing an extra edge distance (equal to the tolerances stack-up) to the part that does not have pilot holes.
- Removable joints: This can be achieved using slotted holes at one of the joining parts.

Polit Holes on Coupling



e: edge distance. $2D+1$ mm

t: assembly tolerances + manufacturing tolerances

Tolerances and manufacturing process: shimming

One of the consequences of manufacturing tolerances are the gaps between the parts to be joined, these gaps can be filled with the following restrictions:

Gap	Shimming
$< 0.3^* \text{ mm}$	No shimming required, sealing depending of drawing requirement
$0.3^* < \text{Gap} < 0.5^*$	Liquid shim
$0.5^* < \text{Gap} < 1.0^*$	Solid shim + liquid shim if gap is not uniform

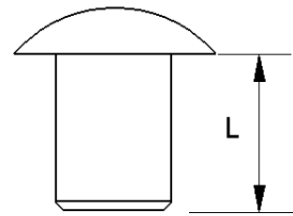
The gaps intervals defined here can be slightly different depending of the project.

Fasteners Installation

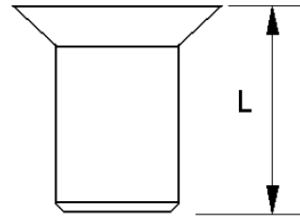
It is necessary to know which is the assembly process to be used and which are the tools available in order to design a joint which allows access to the fasteners not only with the fasteners installation tools but also with the drilling tools when the holes are performed at assembly.

Sometimes the assembly step where a fasteners has to be installed adds restrictions to the kind of fasteners to be used because there is not access from one side of the structure.

Fasteners Installation: Solid rivets

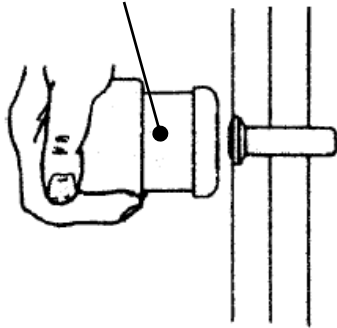


Protruding Rivet

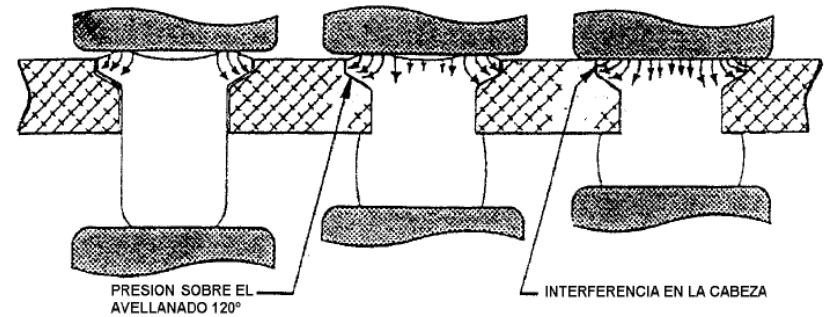
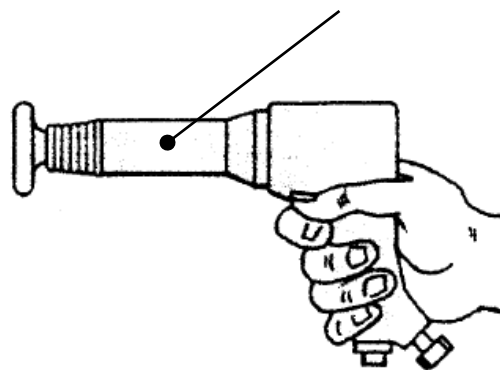


Countersunk Rivet

Backup tool



Head forming tool



Installation and head forming process

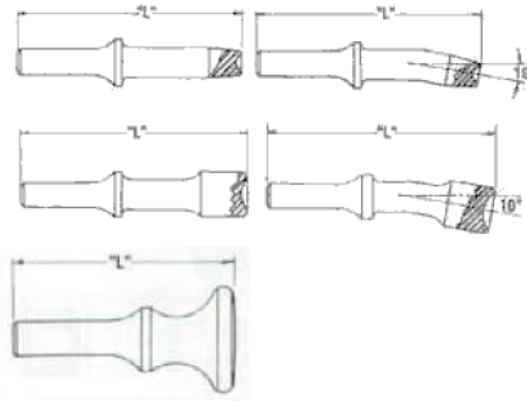
Fasteners Installation: Solid rivets

Riveting technology:

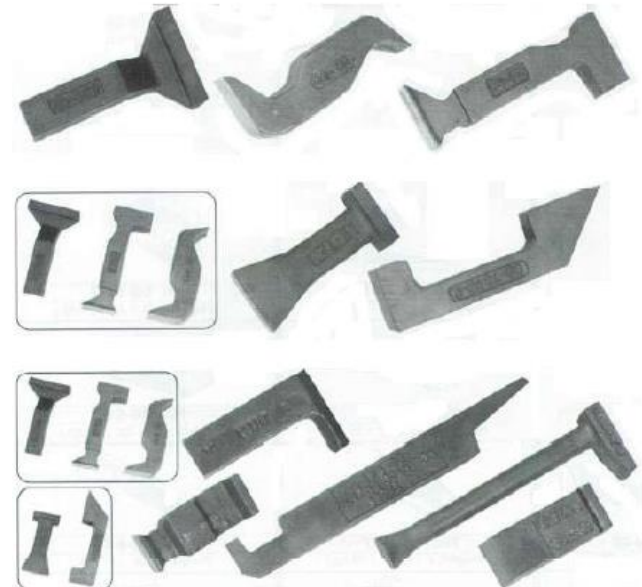
-Automatic installation by pneumatic hammer, head forming tool and backup



Hammer

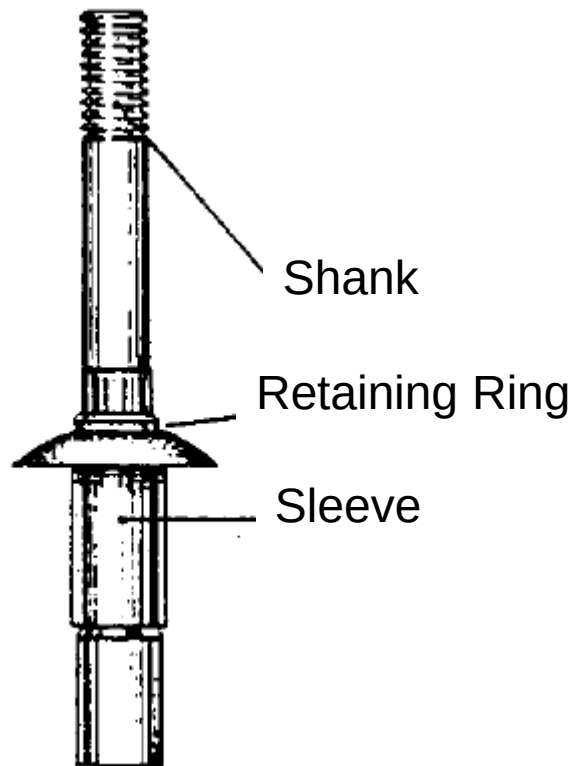


Head forming tool

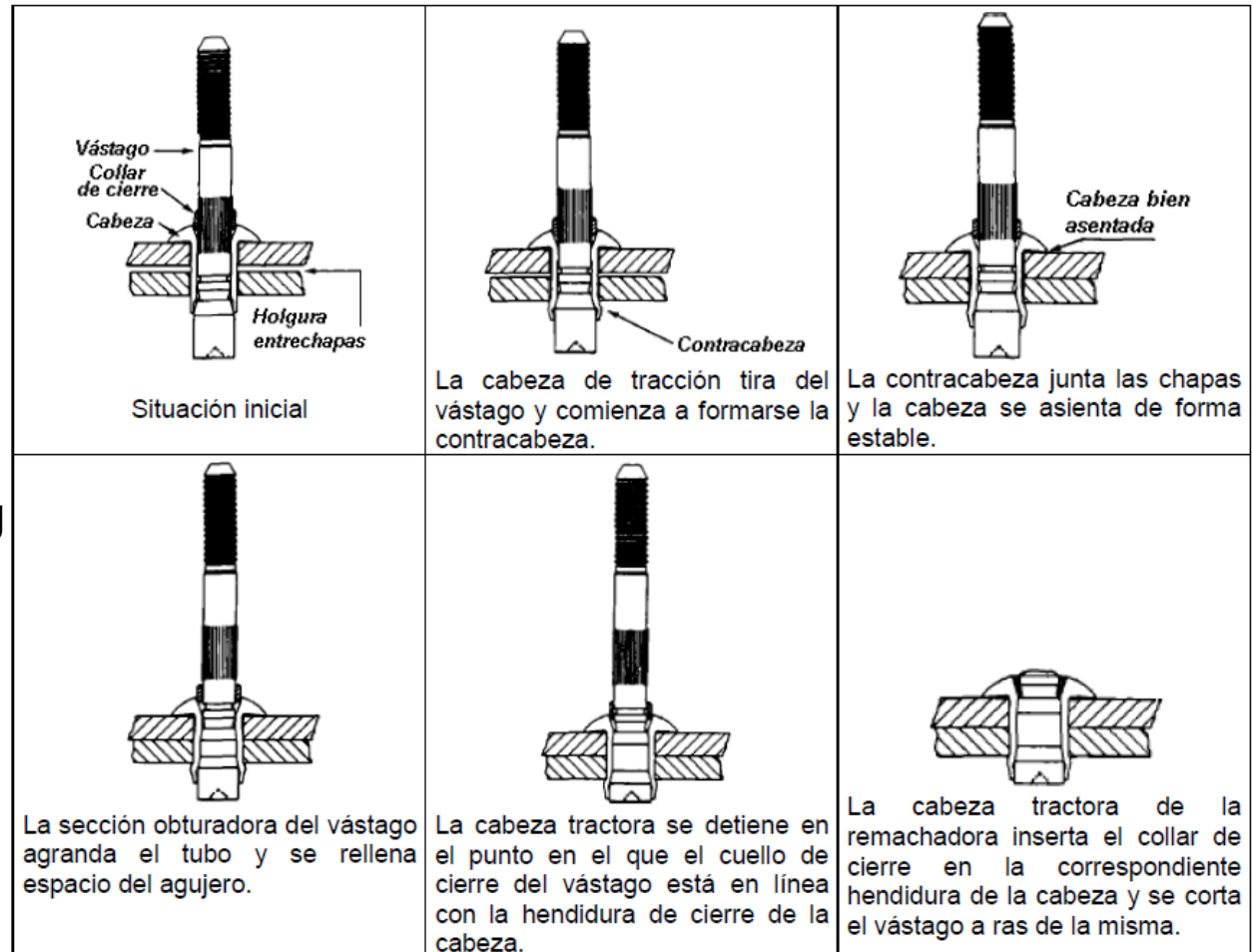


Backup tool

Fasteners Installation: Blind rivets



Cherry-lock



Fasteners Installation: Hi-lok



Protruding Pin

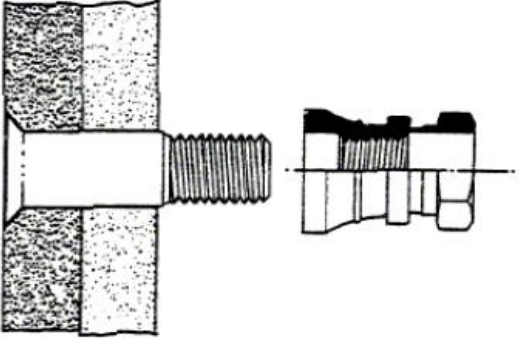
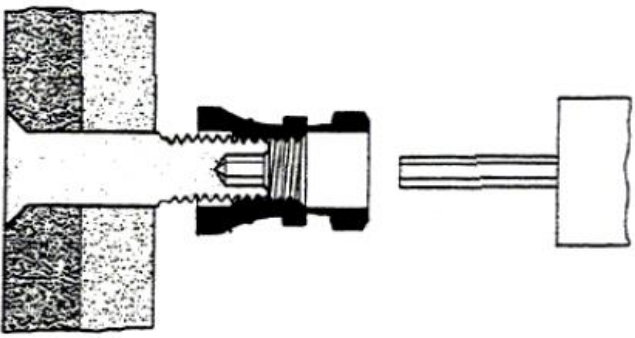
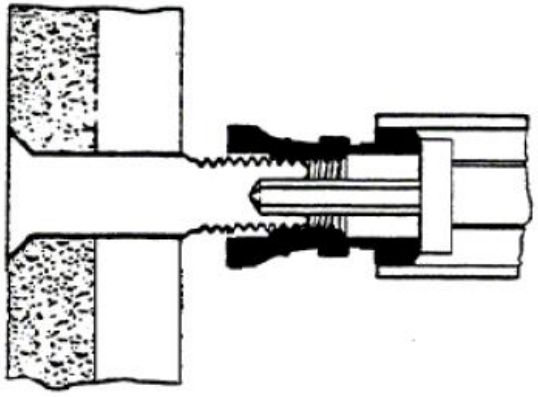
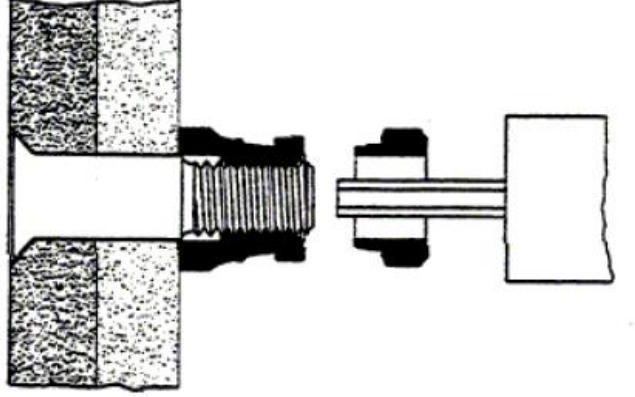


Countersunk Pin

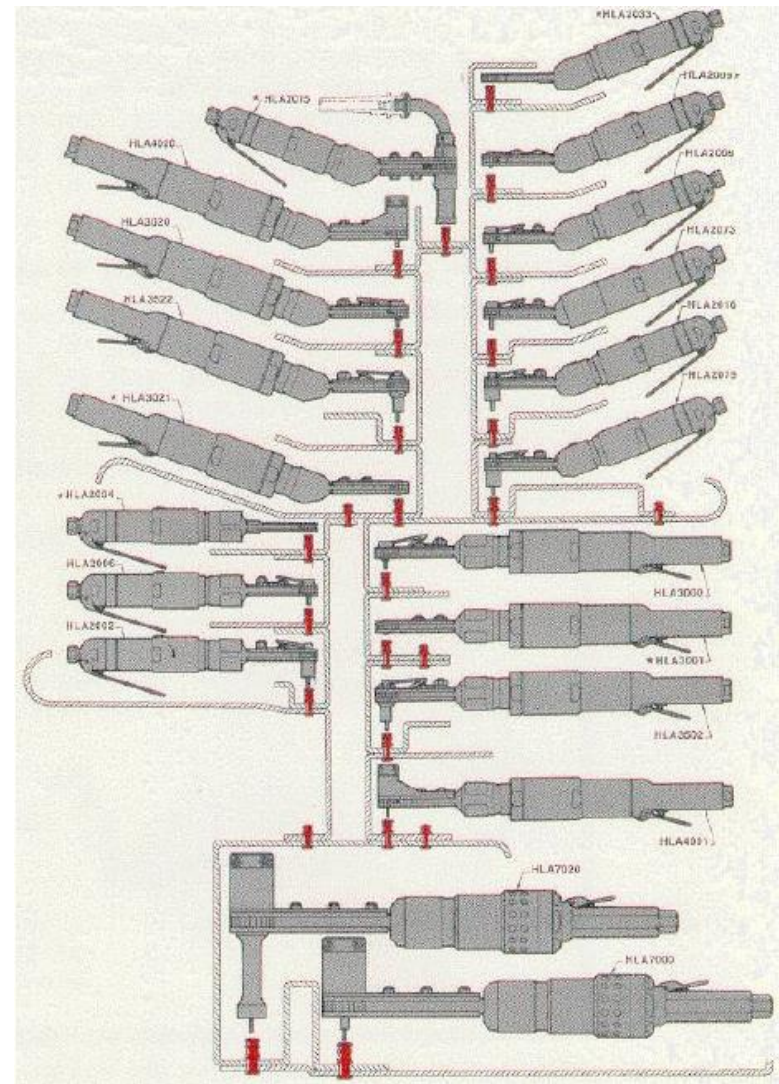
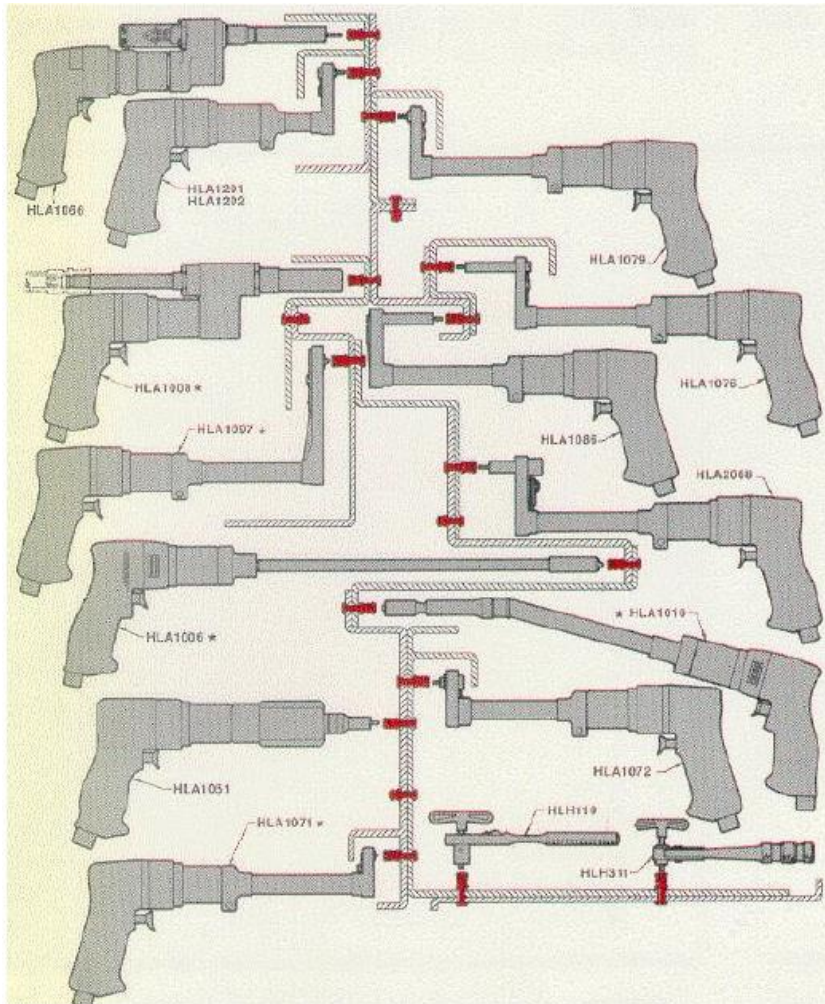


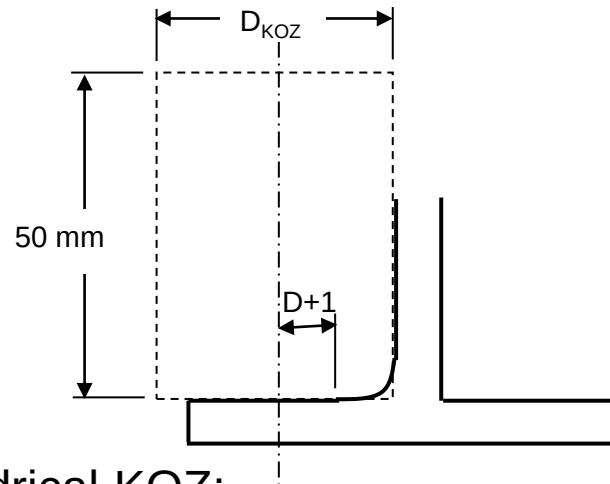
Collar

Fasteners Installation: Hi-lok

	
1. La instalación del perno se completa por uno de los lados de la unión en cuanto se inserta en su agujero correspondiente.	2. En pernos instalados con poca o nula interferencia se introduce en el alojamiento hexagonal del perno una llave allen que evite su rotación.
	
3. A medida que se rosca el collar va aumentando el par de apriete.	4. En el momento de alcanzarse el par de apriete de diseño la tuerca hexagonal del collar se desprende por la línea de corte

Fasteners Installation: Hi-lok



Fasteners Installation: Hi-lok and rivets Keep Out Zone (KOZ)

Hi-lok require a cylindrical KOZ:

- Fasteners D4 to 6.4 mm: D20-22xH50-60
- Fasteners D7.9 to 9.5 mm: D30xH50

This KOZ is required at collar side and at drilling side of the joint.

It is needed a flat circular area around fastener axis of radius $D+1$ in order to guarantee a flat contact surface for the fastener head and the nut/collar.

Blind rivets have specific assembly distances defined at rivet standard or rivet installation standard

Fasteners Installation: Holes diameters

- Interference fit (A): For hi-lok installation in fatigue critical aluminium structures. Interference from 13 to 140 μm
- General Interference fit (AA): Hi-lok typical installation fit. Interference from 0 to 75 μm
- Precision fit (C). Hi-lok installation on CFRP structures. Gap from 0 to 51 μm
- Low clearance fit (E): Jo-bolt / blind bolt installation. Gap from 0 to 77 μm .
- Low clearance fit (EE): Blind rivets installation. Gap from 24 to 235 μm .
- High clearance fit (F): Solid rivets installation. Gap from 40 to 300 μm .
- Mid clearance fit (FF): Solid rivets installation at fatigue critical structures . Gap from 30 to 200 μm .

Corrosion Prevention and Electrical Bonding

The different areas of an aircraft are classified from different points of view, the environmental criticism in terms of corrosion, the susceptibility to receive a lighting strike or be a path for a lighting from the impact zone to the exit zone, the presence of volatile gases which can lead to combustion in case of sparks, etc

The corrosion prevention leads to the selection of specific fasteners materials or surface protection and the use of sealant and paint in order to reduce/avoid the structure degradation due to the corrosion.

The electrical bonding leads to the use of specific fasteners material/surface protections and the use of special bonding washers. As a general rule every metallic part should have electrical continuity with the surrounding metallic parts in order to avoid sparks due to static electricity accumulation.

The electrical bonding of a lighting path usually requires the use of bonding leads or bonding straps in order to guarantee the minimum cross section for the electrical current.

Corrosion Prevention and Electrical Bonding: Protection at assembly

ENVIRONMENTAL CATEGORY	FASTENER MATERIAL	AL OR ST				TI				CRES OR NI				
	STRUCTURE MATERIAL	AL OR AL-FML	TI OR CRES OR CFRP	ST	GRRP OR AFRP	AL OR AL-FML	TI OR CRES OR CFRP	ST	GRRP OR AFRP	AL OR AL-FML	TI OR CRES	ST	CFRP	GFRP OR AFRP
A		2		2	2	2	1	2	1	2	1	2	1	1
B		4		4	4	4	1	4	1	4	1	4	1	1
B (tank boundaries)		6				6	6			6	6		6	
C2-1		3		3	3	3	1	3	1	3	1	3	1	1
C2-2 Also external areas hidden by fairings		8		8	8	8	1	8	1	8	1	8	1 for Cres 8 for Ni	1
External Areas, not hidden by fairings (no classification into categories)		1 before application of final paint system												

GENERIC

ENVIRONMENTAL CATEGORY	PREFORMED HEAD		COLLAR			
	TI OR CRES BOLT		AL	CRES*OR TI*	CRES	TI
	ON AL SIDE OF JOINT	ON CFRP SIDE OF JOINT	ON AL SIDE OF JOINT		ON CFRP SIDE OF JOINT	
A	7	1	3	7	1	1
B (internal)	9	1	9	9	9	1
B (boundary)	12	6	12	12	12	6
C1						
C2-1	8	1	3	3	3	1
C2-2	5	5	10	5	10	11

AL / CFRP JOINTS

* Depending on collar surface protection additional washer might be necessary.

In case of Al/CFRP assemblies in cat. C areas, the re-preservation of the fasteners by varnish only is not permissible; primer or sealant always needs to be applied first.

Corrosion Prevention and Electrical Bonding: Protection at Assembly

Re-preservation schemes

1	No preservation
2	P or Cidap or Varnish
3	P + Tc or P + Varnish or Cidap
4	P or P + Tc or Sealant
5	P + Tc or P + Cidap or P + Varnish or Sealant + P + Tc or Sealant + P + Cidap or Sealant + P + Varnish
6	Sealant overcoating

7	P + Tc or P + Varnish or Varnish or Cidap
8	P + Tc or P + Varnish or P + Cidap
9	P or P + Tc
10	Sealant + P + Tc or Sealant + Varnish or Sealant + Cidap
11	Sealant + P + Tc or Sealant + P + Varnish or Sealant + Varnish or Sealant + Cidap
12	Sealant + P or Sealant + P + Tc

Corrosion Prevention and Electrical Bonding: Bolt/structure

Bolts (parts, which may go through the hole of a part like bolts, rivets, bearings, bushes, inserts)			Structure substrate								
			Al	Ti	Cres	St	CFRP	GFRP AFRP Plastics	Cu	Ni	
Substrate	Ti	Finish	none		✓	✓		✓	✓	✓	✓
			RBC	✓	✓	✓	✓	✓	✓	✓	✓
			SAA	✓	✓	✓	✓	✓	✓	✓	✓
			WC-C:H		✓	✓		not assessed	✓	✓	not assessed
			CuNiIn		✓	✓		not assessed	✓	✓	✓
			IVD-Al	✓	2)	2)	2)		✓	2)	2)
			Ph-Fluorid	✓	✓	✓		✓	✓	✓	✓
	Cres		none		✓	✓		✓	✓	✓	✓
			Passivated		✓	✓		✓	✓	✓	✓
			RBC	✓	✓	✓	✓	✓	✓	✓	✓
			IVD-Al	✓	2)	2)	2)		✓	2)	2)
			Zn-plated 1)	✓			✓		✓	✓	
			ZnNi-plated	✓			✓		✓		
			WC-C:H		✓	✓		not assessed	✓	✓	not assessed
	St		Hard-Cr	not assessed	✓	✓		✓	✓	✓	✓
			Cd-plated	✓	3) 5)		4)	5)	✓		5)
			Cd-plated	✓	3)	4)	✓		✓		
			Hard-Cr	not assessed	✓	✓		✓	✓	✓	✓
			ZnNi-plating	✓		✓	✓		✓		
			Zn-plated 1)	✓			✓		✓	✓	
			Al	CCC	✓					✓	
	Anod			✓					✓		
	Ni-base		none		✓	✓		✓	✓	✓	✓
			RBC	✓	✓	✓	✓	✓	✓	✓	✓
			IVD-Al	✓	2)	2)	2)		✓	2)	2)
			Passivated		✓	✓		✓	✓	✓	✓
			Cd-plated	✓	3)		✓		✓		

	galvanic incompatible
1)	Zn-plating shall not be used in contact with fuel
2)	Combination to be avoided, because IVD-AI may be dissolved by corrosion
3)	washer recommended
4)	Combination to be avoided, because Cd will corrode
5)	Combination to be checked case by case, because Cd will corrode

Structure part					Substrate													
					Al	Cres		Ti		Carbon or low alloy Steel		Cu	Ni		Conductive plastic	Non-conductive plastic		
					Finish													
					TSA	passivated	Cd-plating	none	Zn-spray 1)	Al-spray	WC-C:H	Cd-plating	Phosphating	Cd-plating	None	Cd-plating	none	none
CCC	WC-CH	Zn-spray 1)	SAA		IVD-Al	CuNiIn	Zn-spray 1)	Bronze-spray	ZnNi-plating	Passivated	Zn-spray 1)	Glass ply						
	Hard-Chrome	Zn-plating 1)					Zn-plating 1)		IVD-Al		ZnNi-plating							
	none 2)	ZnNi-plating					ZnNi-plating		IVD-Al									
		IVD-Al					IVD-Al											
		Al-spray					Al-spray											
Substrate	Ti	Finish	none			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
			RBC			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			IVD-Al			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			SAA	PhosFl		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			SAA+P			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
	Cres		Passivated	none		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			RBC			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			IVD-Al			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			ZnNi-plated	Cd-plating		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			Silver**			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
	St		ZnNi-plated	Zn-plating 1)		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			Cd-plated			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			Silver**			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
	Al		PA, Nylon			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			none	CCC	Anod	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			None	Passivated		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
	Ni-base*		IVD-Al			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			Silver**			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			RBC			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
	Cu		ZnNi-plated			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			

1) Zn-plating shall not be used in contact with fuel

2) after wet or dry blasting or sanding

Corrosion Prevention and Electrical Bonding: Bolt/Nut

Fastener (bolts, rivets, bearings, bushes, nuts, collars, washer)						Substrate																				
						Ti				Cres				St			Al		Ni				Cu		Co	Plastics
						Finish																				
						none	RBC	SAA	IVD-Al	none	RBC	IVD-Al	Zn-plated 1)	Cd-plated	Hard-Cr	PA Nylon	none	Anod	none	RBC	IVD-Al	Cd-plated	none	ZnNi-plated	none	Zn-plated 1)
WC-C:H					Pass			ZnNi-plated	Zn-plated 1)			CCC		Pass				Ni-plated								
	CuNiIn				Silver-plated			Cd-plated	ZnNi-plated					Silver-plated				Silver-plated								
	PhosFI				WC-C:H																					
					Hard-Chrome																					
Fastener (bolts, rivets, bearings, bushes, nuts, collars, washer)	Substrate	Finish	Ti	none	WC-C:H	CuNiIn	PhosFI		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓				
				RBC					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
				SAA					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
				IVD-Al					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
			Cres	None	Passivated	Silver-plated	WC-C:H	Hard-Chrome	✓	✓	✓	✓	2)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
				RBC					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
				IVD-Al					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
				Zn-plated 1)	ZnNi-plated	Cd-plated			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
			St	Cd-plated	Zn-plated 1)	ZnNi-plated			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
				Hard-Chrome					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
				PA, Nylon					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
				none	CCC				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
			Al	Anod					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
				none	Passivated	Silver-plated			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
				RBC					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
				IVD-Al					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
Ni	Cd-plated					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					
	none	Ni-plated	Silver-plated			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					
	ZnNi-plated					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					
	none					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					
Co	none					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					
	Zn-plated 1)					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					
Plastics						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					
	Zn-plated 1)					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					

✓

galvancal compatible

galvanically incompatible

1) Zn-plating shall not be used in contact with fuel

2) Cd-plated St/Cres_pass: Combination to be avoided, because Cd will corrode

Typical distances: edge distance and pitch

The edge distance and pitch minimum distances are required in order to reduce the stress concentration factor of the holes.

The edge distance reduction also reduces the bearing allowable of the joint

(e) Edge distance:

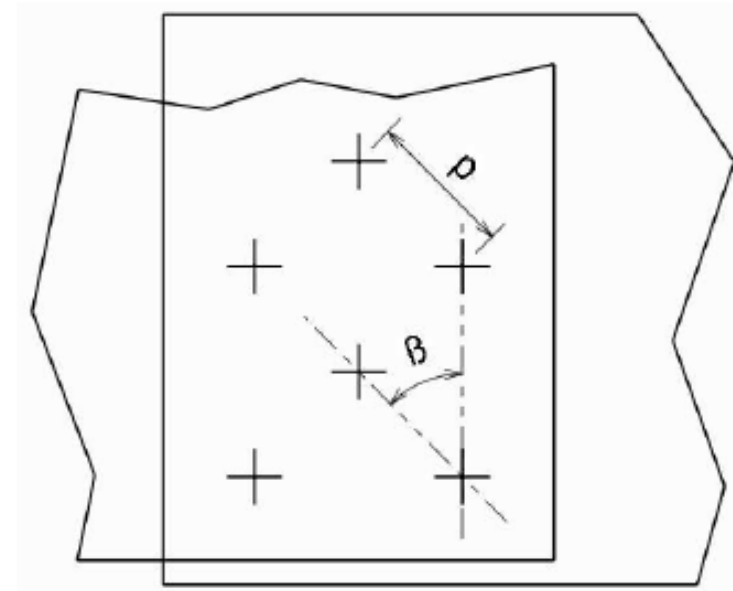
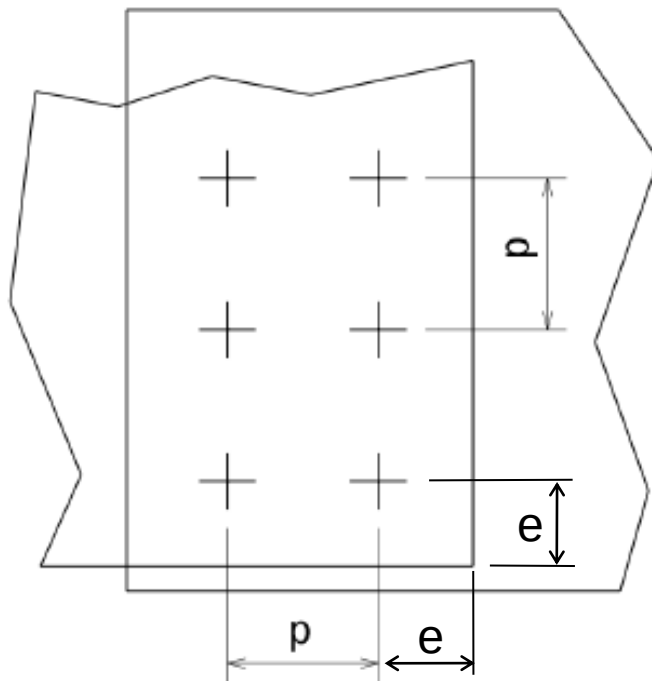
- Metallic Parts: $2D + 1$
- Composite parts:
 - Perpendicular to main load: $2.5D + 1$
 - Aligned to main load: Tension load: $3D + 1$
Compression load: $2.5D + 1$

(p) Pitch:

- Metallic Parts: $4D^*$. Maximum $6D$ (Primary Str.)- $8D$ (Sec. Str.)
- Composite parts: $5D^*$. Maximum 6 (Primary Str.)- $8D$ (Sec. Str.)

*When fasteners diameters are not equal, the mean diameter is considered

Typical distances: edge distance and pitch

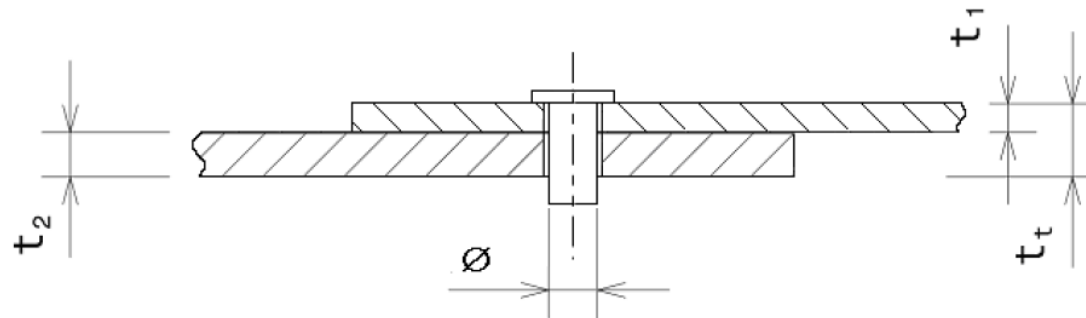


Staggered riveting pattern

β : between 45° and 60°

Stacked thickness:

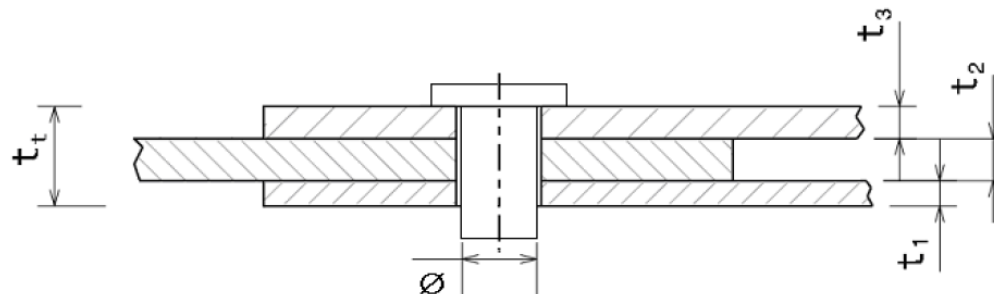
Single Shear:



$$t_t \leq 3 \cdot D$$

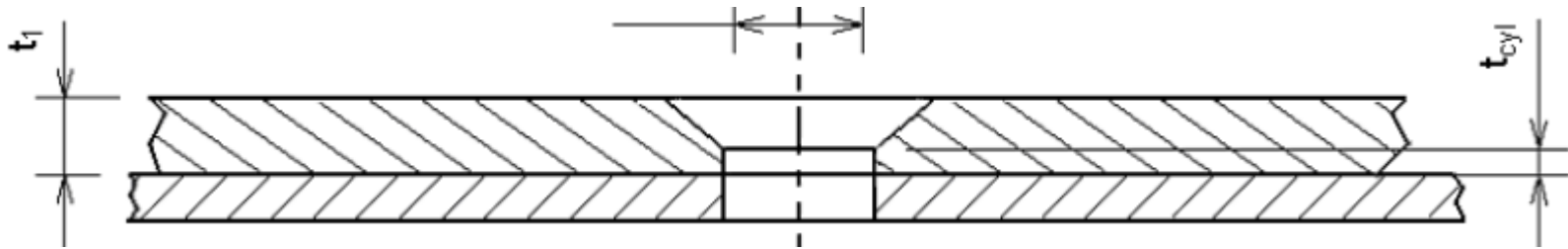
$$t_i \leq 1.7 \cdot D$$

Double Shear:



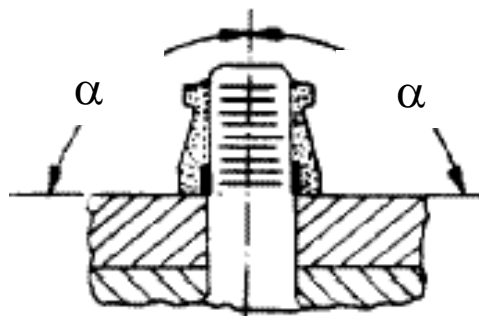
$$t_t \leq 4.5 \cdot D$$

$$t_i \leq 1.7 \cdot D$$

Assembly criteria: Countersunk Fastener (Knife edge)

$$T_{cyl} \geq 0.4\text{mm}$$
$$T_{cyl} \geq 0.3 \cdot t_1$$

These requirements have to be met in order to avoid the knife edge effect which could lead to fastener failure

Assembly criteria: Sloped surfaces

Sometimes the design leads to a misalignment between the joint faces.

The bolt has to be installed at the side where the hole axis is perpendicular to the surface, the nut/collar can be installed at sloped surfaces with the following limits:

Standard	Maximum Slope Allowed
Nut/collar without self-aligning washer	1°
Self-aligning nut/collar	7°
Swage collar	5°
Blind Bolt (ABS025X)	7°

General Criteria

- Reduce to minimum the number of fasteners per joint.
- Use, if possible, the same type and diameter of fastener at one joint.
- Thread must not lie in the joined parts, it must start in the non threaded area of the nut or collar.
- Use washer under bolts head and under nut to avoid parts damage due to torque tighten.

Selection Criteria summary:

	Maintenability	Tension	Shear	Cost
Solid Rivet	Access from both sides	NO	OK	Low
Blind Rivet	Accessibility from one side only	NO	OK	Medium
Hi-lok	Access from both sides	OK	OK	High
Jo-bolt	Acces from one side only	OK	OK	High
Bolt	Acces from one side only	OK	Ok but less efficient due to clearance	Low

1. NT-0-ANC-09015 “Estandares aplicables a C295 CN235 y C212”
2. TAE-T-NT-150063 “FT4B Design HTP & VTP Structure”
3. CAN40027 “Taladros para remaches no métricos”
4. Antonio E. Jimenez Gahete, “Curso de Estructuras en Materiales Compuestos”, Feb 2012
5. ABD0078 “Corrosion Prevention”, Issue E, Dec 2016